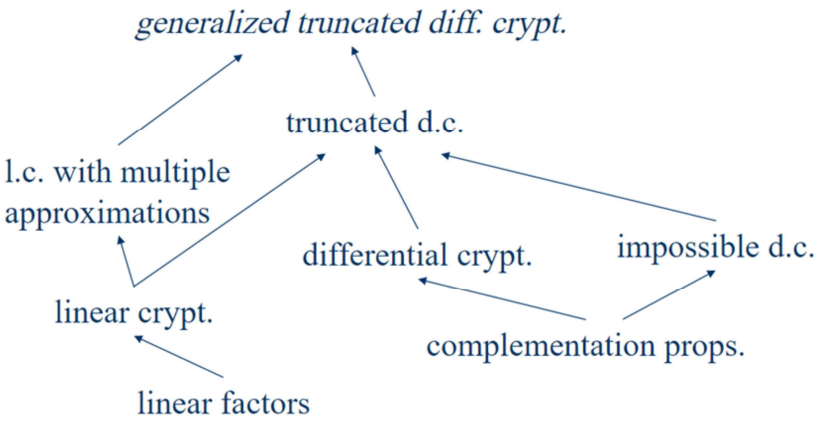
Algebraic Attacks



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Algebraic

Algebraic attacks are a class of techniques that rely for their success on block ciphers exhibiting a high degree of mathematical structure. For instance, it is conceivable that a block cipher might exhibit a group structure. If this were the case, it would then mean that encrypting a plaintext under one key and then encrypting the result under another key would always be equivalent to single encryption under some other single key. If so, then the block cipher would be considerably weaker, and tile use of multiple encryption cycles would offer no additional security over single encryption.

Rainbow Table

User	Password
Stephen	auhsoJ
Lisa	hsifdrowS
James	1010NO1Z
Harry	sinocarD tupaC
Sarah	auhsoJ

User	Password Hash
Stephen	39e717cd3f5c4be78d97090c69f4e655
Lisa	f567c40623df407ba980bfad6dff5982
James	711f1f88006a48859616c3a5cbcc0377
Harry	fb74376102a049b9a7c5529784763c53
Sarah	39e717cd3f5c4be78d97090c69f4e655

User	Random Salt	Password Hash
Stephen	06917d7ed65c466fa180a6fb62313ab9	b65578786e544b6da70c3a9856cdb750
Lisa	51f2e43105164729bb46e7f20091adf8	2964e639aa7d457c8ec0358756cbffd9
James	fea659115b7541479c1f956a59f7ad2f	dd9e4cd20f134dda87f6ac771c48616f
Harry	30ebf72072134f1bb40faa8949db6e85	204767673a8d4fa9a7542ebc3eceb3a2
Sarah	711f51082ea84d949f6e3efecf29f270	e3afb27d59a34782b6b4baa0c37e2958

Figure 1. Password and Hash Tables

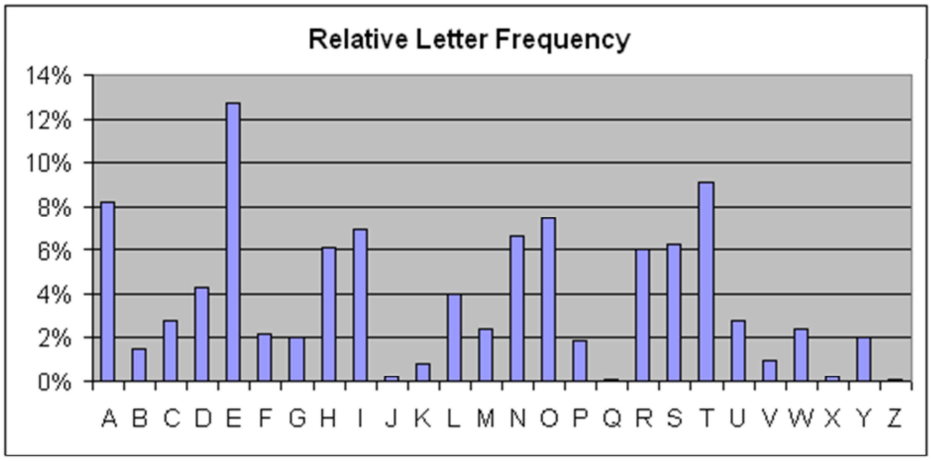
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Rainbow table

Hash functions map plaintext into a hash. Because the hash function is a one-way process, one should not be able to determine the plaintext from the hash itself. To determine a given plaintext from its hash, refer to these two ways to do that:

1. Hash each plaintext until matching hash is found; or
2. Hash each plaintext, but store each generated hash in a table that can used as a look up table so hashes do not need to be generated again. A rainbow table is a lookup table of sorted hash outputs. The idea here is that storing precomputed hash values in a rainbow table that one can later refer to saves time and computer resources when attempting to decipher tile plaintext from its hash value.

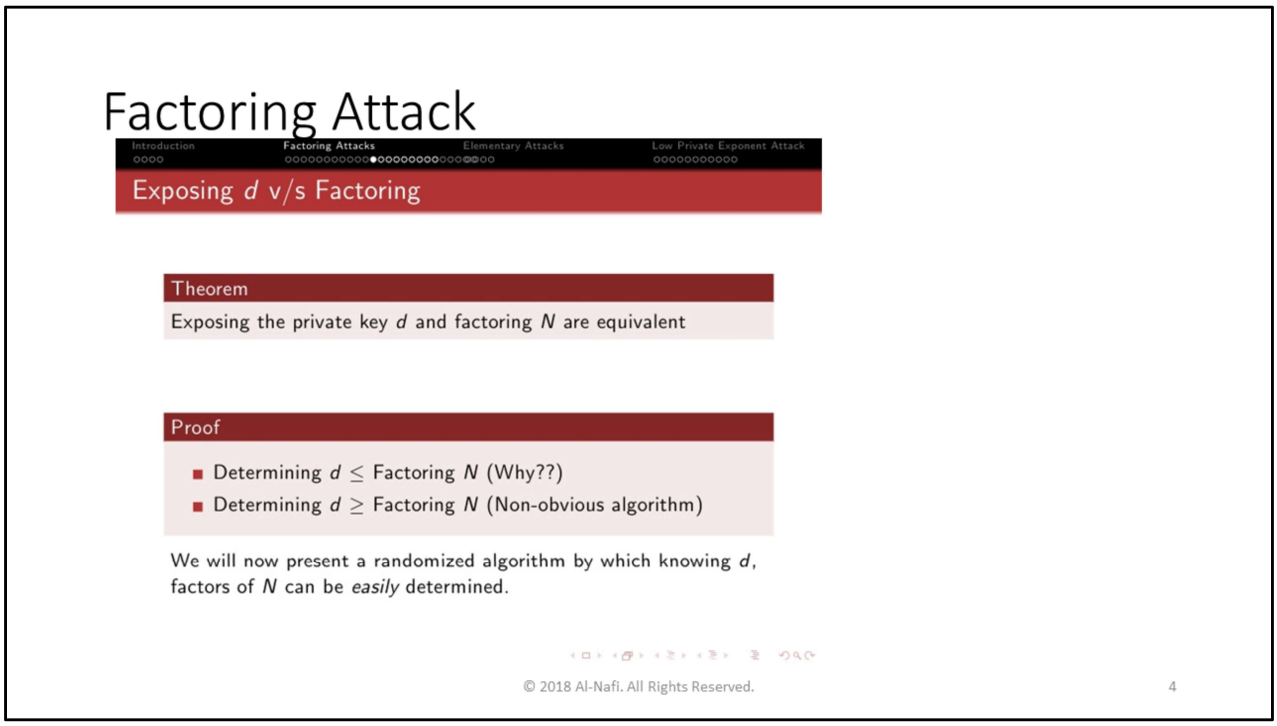
Frequency Analysis



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Frequency analysis

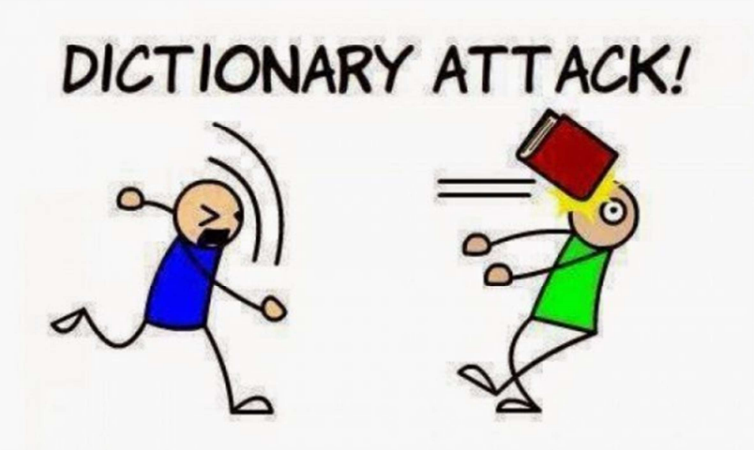
This attack works closely with several other types of attacks. It is especially useful when attacking a substitution cipher where the statistics of the plaintext language are known. In English, for example, some letters will appear more often than others will, allowing an attacker to assume that those letters may represent an E or S.



Factoring Attack

This attack is aimed at the RSA algorithm specifically. Because that algorithm uses the product of large prime numbers to generate the public and private keys, this attack attempts to find the private key through solving the factoring of these public keys.

Dictionary Attack

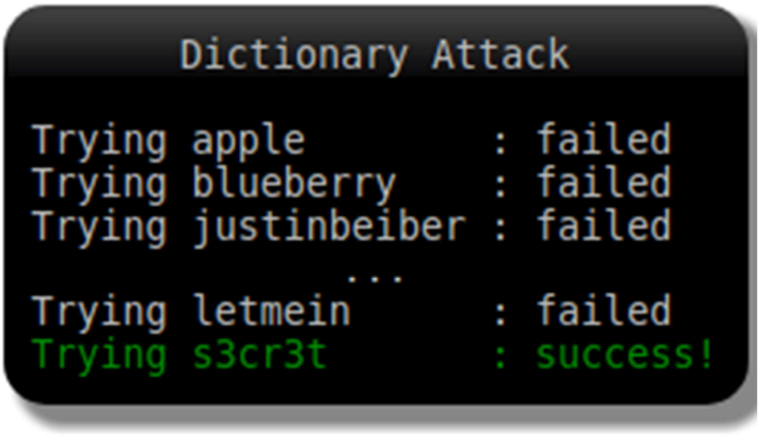


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Dictionary attack

The dictionary attack is used most commonly against password files. It exploits the poor habits of users who choose simple passwords based on natural words. The dictionary attack merely encrypts all of the words in a dictionary and then checks whether the resulting hash matches an encrypted password stored in the SAM file or other password file.

Dictionary Attack Continued…. ☺



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Attacking the Random Number Generators

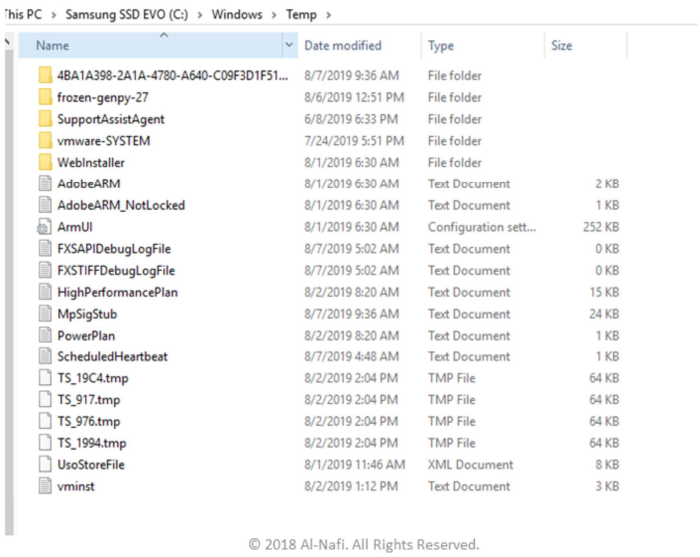
Cryptanalytic Attacks on Random Number Generators

- Examples of random parameters in cryptography:
  - Session keys
  - Numbers to be hashed with passwords
  - Parameters in digital signatures
  - Nonces
- Most of the above are approximated using PRNGs
- For true randomness:
  - Noise in electrical circuits
  - Radioactive decay etc.

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This attack has been successful against certain cryptography implementations. If the random number generator used by
cryptosystems is too predictable, it may give attackers the ability to guess or predict the random numbers that are very critical in setting up initialization vectors in cryptography systems. With this information in hand, the attacker is much more likely to run a successful attack.

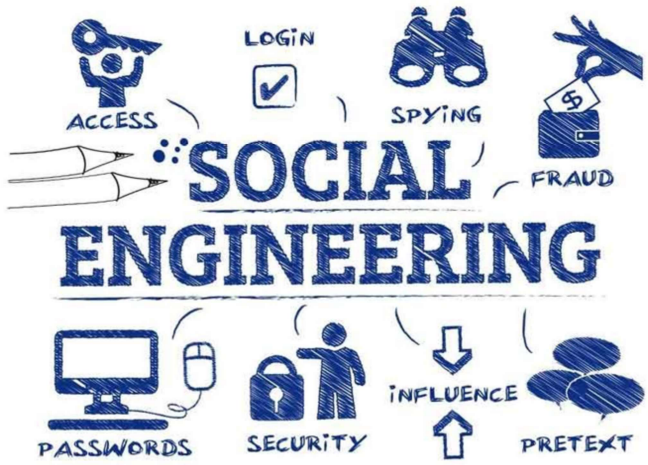
Temporary Files



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Most cryptosystems will use temporary files to perform their calculations. If these files are not deleted and overwritten, they may be compromised and lead an attacker to the message in plaintext.

Social Engineering for Key Discovery



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Social engineering for key discovery

This is the most common type of attack and usually the most successful. All cryptography attacks rely to some extent on humans to implement and operate. Unfortunately, this is one of the greatest vulnerabilities and has led to some of the greatest compromises of a nation's or organization's secrets or intellectual property. Through coercion, bribery or befriending people in positions of responsibility, spies or competitors are able to gain access to systems without having any technical expertise.